lum lookup tables

Purpose

Three lookup tables supply values that `landuse.lum` names by record: `cntable.lum` (SCS curve numbers), `cons_practice.lum` (USLE support practice factor) and `ovn_table.lum` (Manning's n for overland flow). They are named third, fourth and fifth on the `lum` line of `file.cio`, after `landuse.lum` and `management.sch`. If a file is missing or named `null`, every land use gets the built-in values given below.

All three files have the same layout. Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one record, read by position. Values after the last numeric column (the descriptions the SWAT+ Editor writes) are not read. Record names are up to 40 characters and are matched case-sensitively.

cntable.lum

Curve numbers for moisture condition II, one value per hydrologic soil group, for each combination of land cover, treatment and hydrologic condition. The `cn2` column of `landuse.lum` names a record.

#	Column	Units	Built-in value	Range	Description
1	`name`	-	-	-	Curve number record name.
2	`cn_a`	-	30	30 to 100	Curve number for hydrologic soil group A.
3	`cn_b`	-	55	30 to 100	Curve number for hydrologic soil group B.
4	`cn_c`	-	70	30 to 100	Curve number for hydrologic soil group C.
5	`cn_d`	-	77	30 to 100	Curve number for hydrologic soil group D.

The SWAT+ Editor also writes `description`, `treat` and `cond_cov` columns, which are not read. The built-in values apply to land uses whose `cn2` is `null` or not found.

Curve number in the model

At the start of the run each HRU takes the value for the hydrologic soil group (`hyd_grp`) of its soil in `soils.sol`. The table values are those of SCS Engineering Division (1986), which apply to a 5% slope; SWAT+ 62.0.1 does not adjust them for the HRU slope. From CN_2 SWAT+ derives the dry and wet condition values

$$CN_1 = CN_2 - \frac{20\,(100 - CN_2)}{(100 - CN_2) + \exp[2.533 - 0.0636\,(100 - CN_2)]}, \qquad CN_1 \geq 0.4\,CN_2$$

$$CN_3 = CN_2\,\exp[0.006729\,(100 - CN_2)]$$

and the retention parameter curve used for daily runoff, as described in hydrology.

CN_2 can change during the run through the management operation `cnup` in `management.sch`, the decision table action `cn_update` (both limited to 35 to 95), a land use change (`lu_change`, which picks the new land use's record) and the calibration parameter `cn2` in `calibration.cal`.

Example

`refdata/Ames_sub1/cntable.lum` in the SWAT+ repository, first records, with the spacing reduced:

```
cntable.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name        cn_a      cn_b      cn_c      cn_d  description  treat              cond_cov
fal_bare    77.00000  86.00000  91.00000  94.00000  Fallow   Bare_soil          ----
fal_res_p   76.00000  85.00000  90.00000  93.00000  Fallow   Crop_residue_cover  Poor
fal_res_g   74.00000  83.00000  88.00000  90.00000  Fallow   Crop_residue_cover  Good
rc_strow_p  72.00000  81.00000  88.00000  91.00000  Row_crops  Straight_row     Poor
```

The names combine land cover, treatment and condition: `rc_strow_p` is row crops, straight row, poor condition. The SWAT+ Editor's table has 52 records, from fallow and row crops to forest, farmsteads and roads.

cons_practice.lum

USLE support practice factors for conservation practices. The `cons_prac` column of `landuse.lum` names a record.

#	Column	Units	Built-in value	Range	Description
1	`name`	-	-	-	Conservation practice name.
2	`usle_p`	-	1.0	0 to 1	USLE support practice factor P.
3	`slp_len_max`	m	1.0	-	Maximum slope length for the practice. Read but not used (see below).

The SWAT+ Editor also writes a `description` column, which is not read. The built-in values apply to land uses whose `cons_prac` is `null` or not found, so a P factor of 1 means no support practice.

P factor in the model

P enters the MUSLE sediment yield through a multiplier fixed at the start of the run:

$$K \cdot C_{frg} \cdot P \cdot LS \cdot 11.8$$

where K and C_{frg} come from the top soil layer (see soils.sol), and the slope length and steepness factor is

$$LS = \left(\frac{L}{22.128}\right)^{m} \left(65.41 \, \sin^2 \alpha + 4.56 \, \sin \alpha + 0.065\right), \qquad m = 0.6 \left[1 - \exp(-35.835 \, S)\right]$$

with L the slope length (`slp_len` in `topography.hyd`, m), S the slope (m/m) and $\alpha = \arctan S$. The model works out the smaller of L and `slp_len_max`, but then computes LS from L, so `slp_len_max` has no effect on LS in 62.0.1. See erosion and sediment.

P can also be set during the run by the decision table actions `p_factor`, `contour`, `stripcrop` and `terrace`, which recompute the multiplier. The calibration parameter `usle_p` in `calibration.cal` changes the HRU's P factor after the multiplier has been computed and does not recompute it, so by itself it does not change sediment yield.

Example

`refdata/Ames_sub1/cons_practice.lum`, first records:

```
cons_practice.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                    usle_p   slp_len_max  description
up_down_slope          1.00000     121.00000  Up_and_down_slope
cross_slope            0.75000     121.00000  Cross_slope_tillage
contour_farming        0.50000     121.00000  Contour_tillage
strip_cros_slope       0.37000     121.00000  Strip_cropping_cross_slope
```

The SWAT+ Editor's table has 38 records: up-and-down and cross-slope tillage, contouring and strip cropping by slope class (1-2% to 21-25%, with pasture or row crops), and terraces by slope class with sod or underflow outlets.

ovn_table.lum

Manning's roughness coefficient for overland flow by tillage, residue and land cover. The `ov_mann` column of `landuse.lum` names a record.

#	Column	Units	Built-in value	Description
1	`name`	-	-	Manning's n class name.
2	`ovn_mean`	s/m1/3	0.5	Mean Manning's n for overland flow. This is the value the model uses.
3	`ovn_min`	s/m1/3	0.5	Minimum Manning's n. Read but not used; a guide for calibration ranges.
4	`ovn_max`	s/m1/3	0.5	Maximum Manning's n. Read but not used; a guide for calibration ranges.

The SWAT+ Editor also writes a `description` column, which is not read. A land use whose `ov_mann` is `null` or not found does not take a value from this file; its HRUs keep the model's starting value of 0.05.

Manning's n in the model

The HRU's Manning's n sets the overland part of its time of concentration (hours), which controls the peak runoff rate:

$$t_{ov} = \frac{0.0556\,(L\,n)^{0.6}}{(S + 0.0001)^{0.3}}$$

with L the slope length (m) and S the slope (m/m). The same n is used as the roughness of the HRU's concentrated flow channel in the channel part of the time of concentration, and in the area-weighted roughness of routing units.

The time of concentration is computed once, before calibration is applied. The calibration parameter `ovn` in `calibration.cal` changes the HRU's Manning's n after that, and a land use change (`lu_change`) also leaves the time of concentration unchanged, so neither alters the peak runoff rate in 62.0.1.

Example

`refdata/Ames_sub1/ovn_table.lum`, first records:

```
ovn_table.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                ovn_mean       ovn_min       ovn_max  description
fallow_nores         0.01000       0.00800       0.01200  Fallow_no_residue
convtill_nores       0.09000       0.06000       0.12000  Conventional_tillage_no_residue
convtill_res         0.19000       0.16000       0.22000  Conventional_tillage_residue
chisplow_nores       0.09000       0.06000       0.12000  Chisel_plow_no_residue
```

The SWAT+ Editor's table has 20 records, from fallow (0.01) and tilled cropland through range, grasses and forest (up to 0.80 for heavy forest) to urban surfaces (0.011 for asphalt).

Related files

- `landuse.lum`: names records in these tables in its `cn2`, `cons_prac` and `ov_mann` columns.
- `soils.sol`: the hydrologic soil group that picks the curve number, and the K factor and rock content of the MUSLE multiplier.
- `topography.hyd`: slope and slope length.
- `management.sch` and `lum.dtl`: the `cnup` operation and the `cn_update`, `p_factor`, `contour`, `stripcrop` and `terrace` actions.
- Parameter change files: calibration parameters `cn2`, `usle_p` and `ovn`.

Sources

- Readers: `cntbl_read.f90`, `cons_prac_read.f90`, `overland_n_read.f90`, called from `proc_db.f90`
- Type definitions: `type curvenumber_table`, `type conservation_practice_table` and `type overlandflow_n_table` in `landuse_data_module.f90`
- Name lookup: `landuse_read.f90` (`cn2`, `cons_prac`); `plant_init.f90` (`ov_mann`)
- Curve number: `cn2_init.f90`, `curno.f90`; changes in `mgt_sched.f90` (`cnup`) and `actions.f90` (`cn_update`)
- P factor and LS: `plant_init.f90`, `hydro_init.f90`, `ero_ysed.f90`, `actions.f90`
- Manning's n: `time_conc_init.f90`
- Calibration: `cal_parm_select.f90`; call order in `main.f90.in`
- File name slots: `in_lum%cntable_lum`, `in_lum%cons_prac_lum`, `in_lum%ovn_lum` in `input_file_module.f90`

Column	Fortran variable
`cntable.lum name`	`cn%name`
`cn_a`, `cn_b`, `cn_c`, `cn_d`	`cn%cn(1)` to `cn%cn(4)`, copied to `cn2(j)`
`cons_practice.lum name`	`cons_prac%name`
`usle_p`	`cons_prac%pfac`, copied to `hru%lumv%usle_p`
`slp_len_max`	`cons_prac%sl_len_mx`

Column	Fortran variable
`ovn_table.lum name`	`overland_n%name`
`ovn_mean`	`overland_n%ovn`, copied to `hru%luse%ovn`
`ovn_min`, `ovn_max`	`overland_n%ovn_min`, `overland_n%ovn_max`